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Phosphate speciation in sodium borosilicate glasses studied by nuclear magnetic resonance

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Abstract

The structure of $R\mathrm{Na_2O} \cdot \mathrm{B_2O_3} \cdot K\mathrm{SiO_2} \cdot x\mathrm{P_2O_5}$ ($0.5 < R < 2$; $0.86 < K < 3$) borosilicate glasses has been studied by nuclear magnetic resonance (NMR). ^{31}P magic angle spinning (MAS), double quantum-magic angle spinning (DQ-MAS) and ^{31}P–^{11}B transfer of populations under double resonance magic angle spinning (TRAPDOR MAS) NMR were used to determine the phosphate speciation in the glasses and their connectivity with the borosilicate network. The structure of the glass network was characterized with ^{11}B, ^{29}Si and ^{23}Na MAS NMR. *Ab initio* calculations of the ^{31}P chemical shielding were carried out in order to confirm the connectivity between phosphorus and the structural units of the borosilicate glass network. Na_3PO_4 (monophosphate), $Na_4P_2O_7$ (diphosphate) and P–O–B species (mono- and diphosphate groups with borate units as the next nearest neighbors) are found all along the compositional range studied. The proportion of the P–O–B groups increases as the glass optical basicity decreases, while the proportions of mono- and diphosphate species decrease. The change in the glass transition temperature of the phospho-borosilicate glasses with respect to that of the borosilicate ones is discussed in terms of the structural characterization. The formation of phosphate species gives rise to the increase in T_g, which is attributed to the re-polymerization of the silicate network, while the formation of P–O–B bonds weakens the glass network and produces a decrease in the glass transition temperature.

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Keywords: Glasses; Nuclear magnetic (and quadrupole) resonance; Borosilicates; Phosphates; NMR, MAS-NMR and NQR; Structure; Short-range order

1. Introduction

The importance of borosilicate glasses is demonstrated through their large number and different types of applications. Chemically and mechanically resistant materials constitute one of their most important large-scale industrial applications. Moreover, new and forefront technologies induce the development of materials for which borosilicate glasses are also candidates. In particular, their use as sealing components constitutes one important domain in the current material technology. Conventional glass compositions have been traditionally used as sealants in television tubes or bulb lamps, but the increasing development of microelectronics and new demands, like the worldwide research programs on fuel cells, necessitates new glasses. Recent works have concerned this type of special sealing glasses; in particular, borosilicate glasses have been developed as sealants for Molten carbonate fuel cells (MCFC) [1,2].

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Phospho-silicate and phospho-borosilicate glasses are mainly used in fiber optics technology and lasers [3–5], in microelectronics as insulating interconnections in integrated circuits [6,7] and as protonic conductors in the form of phospho-silicate sol–gel glasses [8,9]. In these cases, despite that phosphorus appears as a minor component, its use improves the properties of glasses. In natural geologic silicate glasses, P_2O_5 affects the repartition of high field strength elements [10] and the redox equilibrium [11]. In silicate and borosilicate glasses, P_2O_5 affects the viscosity [12], the liquidus temperature [12] and the chemical durability [13].

Nuclear magnetic resonance investigations have shown that P_2O_5 in silicate glasses is depolymerized into mono- (PO_4^{3-}) and diphosphates ($P_2O_7^{4-}$), which is counterbalanced by the polymerization of the silicate network and this process depends on the acid–base character of the glass [14]. Phosphorus also influences the devitrification and crystallization of glasses, and it is known as an important nucleating component [15]. P_2O_5 is always present in minor quantities in nuclear waste glass compositions [16], hence the phosphorus speciation and solubility has to be controlled in order to avoid problems with undesirable devitrification phenomena.

The structure of borosilicate glasses has been widely studied [17–20]. It is based on Q^n [SiO_4] tetrahedra, where n is the number of bridging oxygen atoms per tetrahedron, and [BO_3] and [BO_4] units. The [BO_3] units are tricoordinated borons in both ring and non-ring configurations, while the [BO_4] ones are boron atoms in fourfold coordination where oxygen atoms can be bonded to one boron and three silicons (1B,3Si) or four silicon atoms (0B,4Si) and have a Na^+ ion as a charge compensator. The distribution of the boron and silicon structural units depends on the glass composition. Borosilicate glasses are usually defined as a function of the ratios Na_2O/B_2O_3 (R) and SiO_2/B_2O_3 (K) [20]. Dell et al. proposed a structural model for which the variation of [BO_4] borate units (N_4) is represented as a function of the R parameter [20]. For all K values, N_4 increases up to a maximum in between 0.5 and 0.75 for $R = 1$, and the value of this maximum increases with an increasing K parameter. Then, the proportion of N_4 slowly decreases for higher R values. Martens and Müller-Warmuth [21] have also demonstrated that both borate and silicate networks and modifier cations are statistically mixed, with no evidence for distinct compositional regimes.

Rong et al. studied the speciation of phosphorus in aluminoborosilicate glasses by NMR [22]. In all of the compositions studied, they found resonances at 16 and 3.4 ppm, assigned to mono- and diphosphate species, respectively. Another resonance centered at −5 ppm was found, which was assigned to diphosphate units bonded to Al clusters [22]. Furthermore, the authors pointed out that while diphosphate species are the dominant structural units, if monophosphate groups exist, they must be present as isolated droplets within the glass matrix.

The phosphorus speciation in K_2O–B_2O_3–SiO_2–P_2O_5 glasses has been studied with NMR by Gan et al. [23]. They concluded that phosphorus appears as mono- and diphosphate groups, where the latter can be bonded to borate units through P–O–B bonds. As a result of the phosphorus incorporation they observed an increase in the [BO_3]/[BO_4] ratio and an increase in the degree of polymerization of the silicate network.

Although the presence of P–O–B bonds in phospho-borosilicate glasses seems to be demonstrated, there is no consensus with respect of the P–O–Si ones. Cody et al. concluded in the presence of P–O–Si bonds in sodium phosphorus–aluminosilicate glasses for low alumina and modifiers contents [24]. However, no resonances for P–O–Si bond are visible on the NMR spectra of Na_2O–SiO_2–P_2O_5 glasses [13] containing slightly higher modifier content than in the glasses studied by Cody et al. [24]. The only clear evidence for phosphate to silicate bonding was obtained in low alkali glasses [25], or in glasses with a high amount of P_2O_5 [26]. Since there are not enough alkali ions to compensate the charge of phosphate groups, silicon is extracted from the glass network and its coordination changes from tetrahedral to octahedral in order to assume the charge compensation of phosphates [25,26]. A new resonance at −35 ppm then appears on the ^{31}P NMR spectra. So, the localization of a resonance assigned to phosphate connected to tetrahedral silicon (i.e. located in the glass network) in the NMR spectra of silico-phosphate glasses containing a moderate to high alkali content remains unclear.

This work aims to bring further insights on the phosphorus speciation in sodium borosilicate glasses, its evolution with the glass composition, and its influence on the glass transition temperature (T_g). The structure of the glasses is studied by ^{31}P, ^{11}B, ^{29}Si and ^{23}Na NMR. ^{31}P double quantum (DQ) magic angle spinning (MAS) NMR and transfer of populations under double resonance (TRAPDOR) between ^{31}P and ^{11}B nuclei, experiments are also used to probe the connectivity between the phosphates and the glass network. *Ab initio* calculations of the ^{31}P chemical shielding tensors in phosphate model clusters are used to confirm the chemical shift assignment of phosphate and borophosphate species. The evolution of phosphate species with composition is discussed in terms of the glass optical basicity. The evolution of the glass transition temperature (T_g) with the composition of the phospho-borosilicate glasses is analyzed with the results of the structural characterization.

2. Experimental section

The sodium borosilicate and sodium phospho-borosilicate glasses were prepared by melting batches in 90:10 Pt:Rh crucibles during 1 h at a temperature ranging from 1200 to 1550 °C depending on composition, and then quenching over a brass plate. The batches were prepared from reagent grade SiO_2, H_3BO_3, Na_2CO_3 and $(NaPO_3)_n$.

In order to reduce the NMR relaxation time, 0.1 wt% Co_2O_3 was added to the batches.

The glass transition temperature was measured with differential scanning calorimetry at a constant heating rate of 5 K min^{-1}.

^{29}Si MAS NMR spectra were recorded on a 2.35 T spectrometer at a frequency of 19.89 MHz and a spinning rate of 5 kHz. The applied pulse length was 2.3 μs (π/4) and a 15 s recycling time was used. TMS was used as the chemical shift reference. ^{23}Na MAS NMR spectra were recorded at 9.4 T at a frequency of 105.84 MHz and a spinning rate of 12.5 kHz. The pulse length was 2 μs (π/4) and a 5 s recycling time was used. The chemical shift reference used was a 1 M NaCl solution. ^{11}B MAS NMR spectra were recorded at 9.4 T at a frequency of 128.38 MHz and a spinning rate of 10 kHz. The pulse length was 1.5 μs (π/4) and a 3 s recycling time was used. A mixture of BPO_4 and $NaBH_4$ was used as the chemical shift reference. ^{31}P MAS NMR spectra were recorded at 9.4 T at a frequency of 161.92 MHz and a spinning rate of 10 kHz. The pulse length was 2.5 μs (π/4) and a 60 s recycling time was used. The chemical shift reference was a solution of 85% H_3PO_4. ^{31}P DQ-MAS NMR spectra were recorded at 9.4 T at a MAS frequency of 10 kHz. The BaBa (back to back) sequence was applied using TPPI (time proportional phase increment) [27–29]. A 3 μs pulse (π/2) and a 20 s recycling delay were used, with a pre-saturation pulse sequence. In all NMR experiments the recycling delays were chosen to be long enough to enable relaxation. The ^{31}P and ^{11}B NMR spectra were fitted using the DMFIT software [30].

A TRAPDOR experiment was performed between ^{31}P ($I = 1/2$) and ^{11}B ($I = 3/2$) nuclei, at 9.4 T. This experiment combines MAS with rotor-synchronized radio-frequency pulses to probe the dipolar interaction between the two nuclei. A rotor synchronized Hahn echo ($\pi/2 - n\tau_R - \pi - n\tau_R$ − acq) was used to excite ^{31}P nuclei while a RF pulse of 60 kHz was applied on the ^{11}B channel during the first $n\tau_R$ delay of the echo.

Ab initio chemical shielding calculations were performed with the Gaussian03 package [31] using the hybrid HF/DFT method B3LYP [32]. The cluster geometries were optimized with the standard valence double dzeta basis set 6-31G**. NMR shielding tensors are computed using the gauge including atomic orbitals (GIAO) [33] and the NMR-optimized IGLO-III basis set [34].

3. Results

3.1. Glass compositions

In order to obtain a wide and representative sampling of the different structural features in the borosilicate glasses, we have chosen the compositions shown in Fig. 1 to carry out a systematic study. The two lines of compositions have been selected following the structural model of Dell et al. for borosilicate glasses [20]. They include compositions with $0.86 < K < 3$ for $R = 1$ (F-C line) and $0.5 < R < 2$ for $K = 2$ (G-I line). When varying K, for a constant R, the proportion of N_4 boron structural units increases together with an increase in the degree of polymerization of the silica sub-network [20]. On the other hand, an increasing R for a constant K results mainly in an important decrease in the degree of polymerization with smaller variations in the N_4 proportion. The open circles in the ternary diagram represent the C, G and I glass compositions for which the P_2O_5-containing glasses (3 mol%) C3P, G3P and I3P, respectively, showed opalescence. These specimens were amorphous to X-ray diffraction, though their examination by scanning electron microscopy (SEM) revealed the presence of crystallites smaller than 2 μm in the case of the I3P sample, and a droplet microstructure due to phase separation in G3P. However, none of these three samples were used in analyzing the relationship between composition and structure. Besides these three compositions, all of the glasses revealed to be transparent and bubble free.

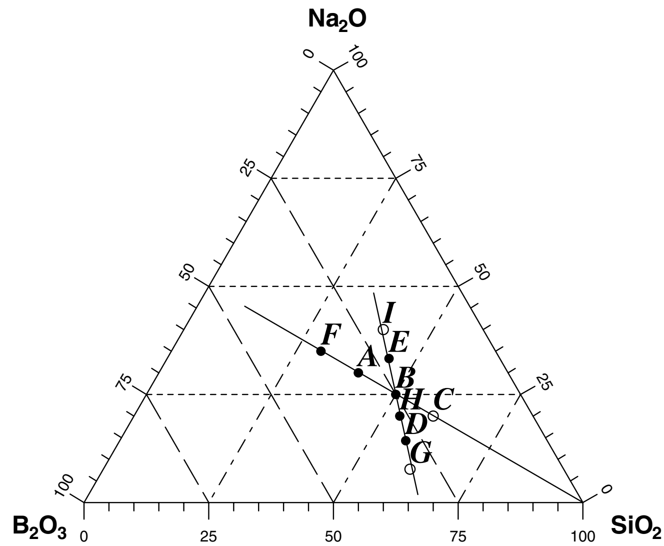

Fig. 1. Na_2O–B_2O_3–SiO_2 ternary diagram showing the glass compositions with $0.86 < K < 3$ and $R = 1$ (F-C line) and $0.5 < R < 2$ and $K = 2$ (G-I line). Open circles represent glass compositions for which the P_2O_5 (3 mol%)-containing glasses showed opalescence.

Table 1 shows the nominal compositions of the borosilicate and phospho-borosilicate glasses prepared, in mol%, the R and K values of the $R\text{Na}_2\text{O} \cdot \text{B}_2\text{O}_3 \cdot K\text{SiO}_2$ borosilicate compositions, the glass optical basicity (Λ_g) calculated with the coefficients of Duffy and Ingram [35], and the T_g values.

3.2. ^{31}P NMR

Fig. 2 presents a series of ^{31}P MAS NMR spectra of the borosilicate glasses containing 3 mol% of P_2O_5. The spectra are organized from top to bottom, by increasing values of the optical basicity of the borosilicate. They all show, although not in the same proportion, three main resonances centered at 15, 3 and −7.5 ppm. The two first reso-

Table 1
Compositions of the borosilicate and phospho-borosilicate glasses, in mol%, the R and K values of the $R\mathrm{Na_2O}\cdot\mathrm{B_2O_3}\cdot K\mathrm{SiO_2}$ borosilicate compositions, the glass basicity (Λ_g) calculated as in Duffy and Ingram [35], and the glass transition temperatures

Sample	R; K	Λ_g	T_g (°C)	Na_2O	B_2O_3	SiO_2	P_2O_5
A	1; 1.3	0.55	492	30	30	40	
A1P	1; 1.3		502	29.7	29.7	40	1
A2P	1; 1.3		507	29.4	29.4	39.2	2
A3P	1; 1.3		502	29.1	29.1	38.8	3
A6P	1,1.3		505	28.2	28.2	37.6	6
B	1; 2	0.54	515	25	25	50	
B1P	1; 2		530	24.75	24.75	49.55	1
B2P	1; 2		510	24.5	24.5	49	2
B3P	1; 2		517	24.25	24.25	48.5	3
B6P	1; 2		503	23.5	23.5	47	6
C	1; 3		n.m.[a]	20	20	60	
C3P	1; 3		n.m.[a]	19.4	19.4	58.2	3
D	0.5; 2	0.50	501	14.3	28.6	57.1	
D3P	0.5; 2		479	13.9	27.7	55.4	3
E	1.5; 2	0.58	480	33.3	22.2	44.5	
E3P	1.5; 2		491	32.3	21.5	43.2	3
F	1; 0.86	0.57	474	35	35	30	
F3P	1; 0.86		490	33.95	33.95	29.1	3
G	0.25; 2	0.48	492	7.7	30.8	61.5	
G3P	0.25; 2		455	7.47	29.88	59.65	3
H	0.75; 2	0.52	563	20	26.7	53.3	
H3P	0.75; 2		537	19.4	25.9	51.7	3
I	2; 2	0.61	429	40	20	40	
I3P	2; 2		(271[b]) 498	38.8	19.4	38.8	3

[a] Not measured.
[b] First endothermic peak.

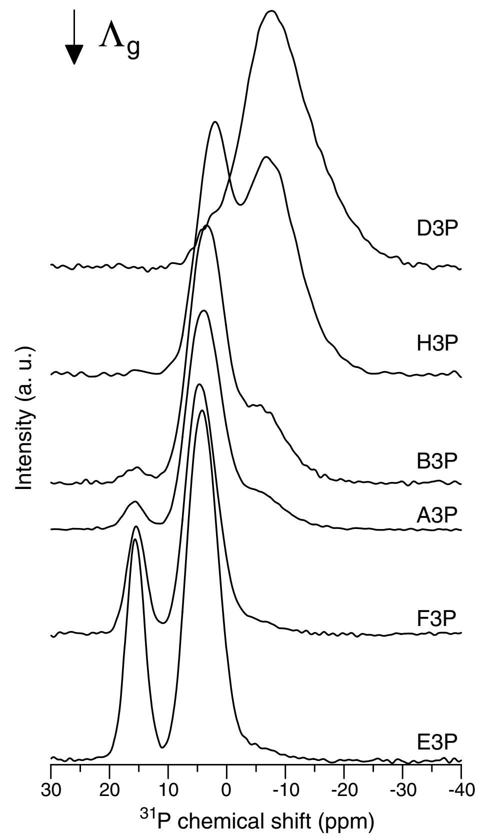

Fig. 2. ^{31}P MAS NMR spectra of borosilicate glasses containing 3 mol% P_2O_5. They are organized, from top to down, by increasing values of the glass optical basicity of the borosilicate glasses.

nances stem from mono- (PO_4^{3-}) and diphosphate ($P_2O_7^{4-}$) species, which also appear in sodium phosphate glasses [36] and in silicate, borosilicate and aluminosilicate glasses [22–24]. The third one is assigned to phosphates bonded to borate units, since they appear in borosilicate glasses and not in silicate glasses [22,23]. They will be named hereafter P–O–B units. Nevertheless, the large chemical shift distribution of this latter resonance suggests that more than a unique type of phosphate-borate connection occur. This question will be addressed in the following using *ab initio* calculations.

Fig. 3 shows the ^{31}P NMR spectra of the borosilicate glass compositions A and B for increasing P_2O_5 contents. While the three phosphate species are present for all of the P_2O_5 contents, a decrease in the amplitude of the mono- and diphosphate resonances, and an increase of that of the P–O–B group occur when the phosphorus content increases.

A ^{31}P DQ-MAS NMR spectrum, shown in Fig. 4, was recorded on the B6P glass. During the evolution time of the DQ experiment, the spin coherences evolve under the effect of dipolar interaction. Thus, it enables to characterize the spatial proximity of nuclei, and it provides information about the connectivities of phosphate sites in glass networks [37,38]. The PO_4^{3-} resonance does not appear in the DQ spectrum (notice that the spectrum scale is centered on $P_2O_7^{4-}$ and B–O–P resonances for clarity), which means that these units are dispersed within the glass network, since dipolar interactions are not reintroduced by the DQ pulse sequence. The $P_2O_7^{4-}$ resonance is located on the diagonal, at 2.5 ppm in the MAS dimension (site 1). Sites located on diagonal of the DQ spectra are self-correlated, meaning that they are close enough to undergo dipolar interaction. This is of course expected for the two PO_4 sites in $P_2O_7^{4-}$ anion. At −8.5 ppm in the MAS dimension, a signal located on the diagonal is assigned to auto-correlated P–O–B phosphate species (site 3). Furthermore, two off-diagonal correlation pairs are found at (3.5; −7) and (0.5; −9) ppm in the MAS dimension, which stem from cross-correlations between the $P_2O_7^{4-}$ and P–O–B groups (sites 2 and 2′). This correlation means that $P_2O_7^{4-}$ species are close to borate units and are probably bonded to each other. ^{31}P–^{11}B heteronuclear through-bond correlation experiment would enable to confirm the presence of P–O–B bonds, but the low P_2O_5 concentration in the samples prevented us from recording such a correlation spectrum.

Further evidence for phosphate and borate proximity was checked through the heteronuclear dipolar ^{31}P–^{11}B interaction, using the TRAPDOR (transfer of population under double resonance) experiment. Fig. 5 shows the ^{31}P–^{11}B TRAPDOR spectrum of the B6P glass. The TRAPDOR effect is revealed by comparing the ^{31}P spectrum obtained without dipolar recoupling (the MAS

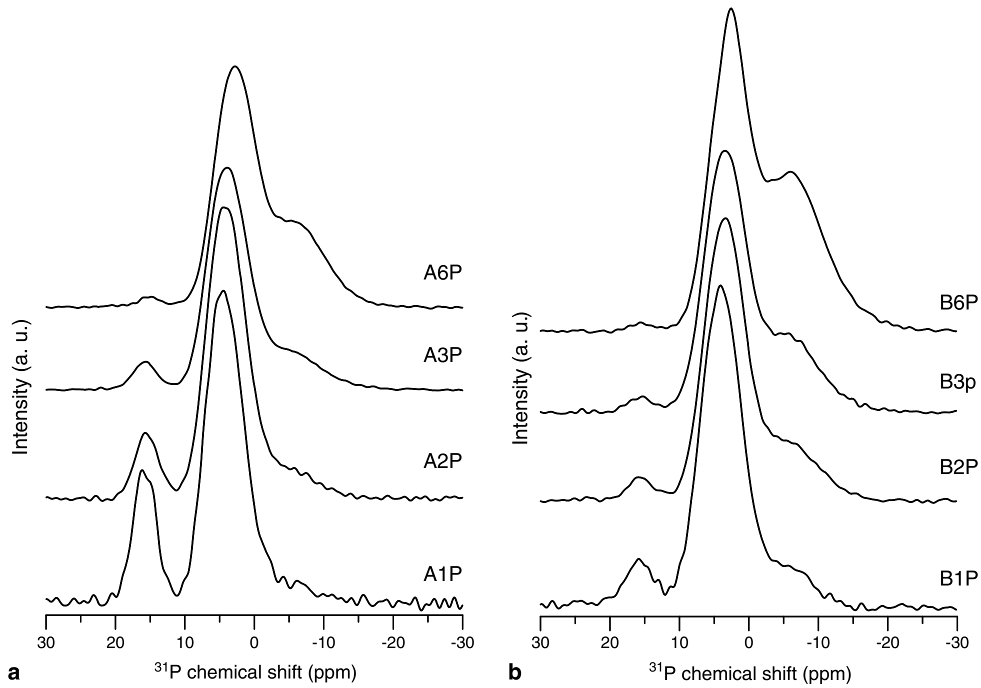

Fig. 3. ^{31}P MAS NMR spectra of the borosilicate glasses A and B, for increasing P_2O_5 contents.

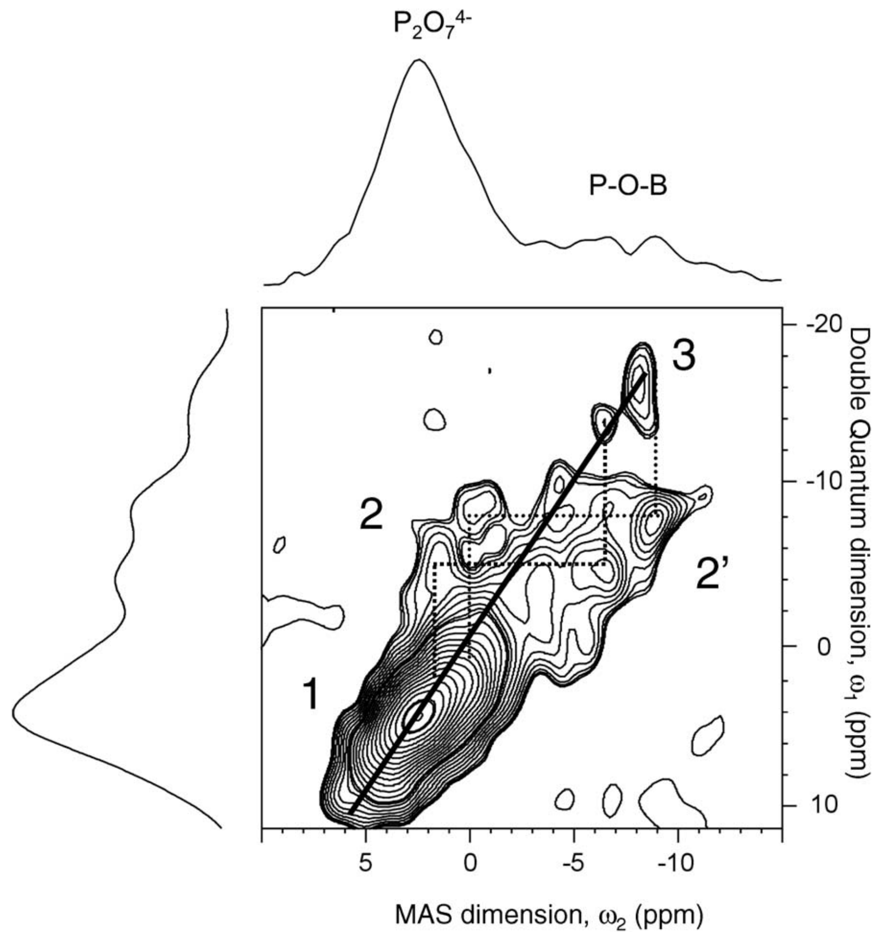

Fig. 4. ^{31}P DQ-MAS NMR spectrum of the glass B6P.

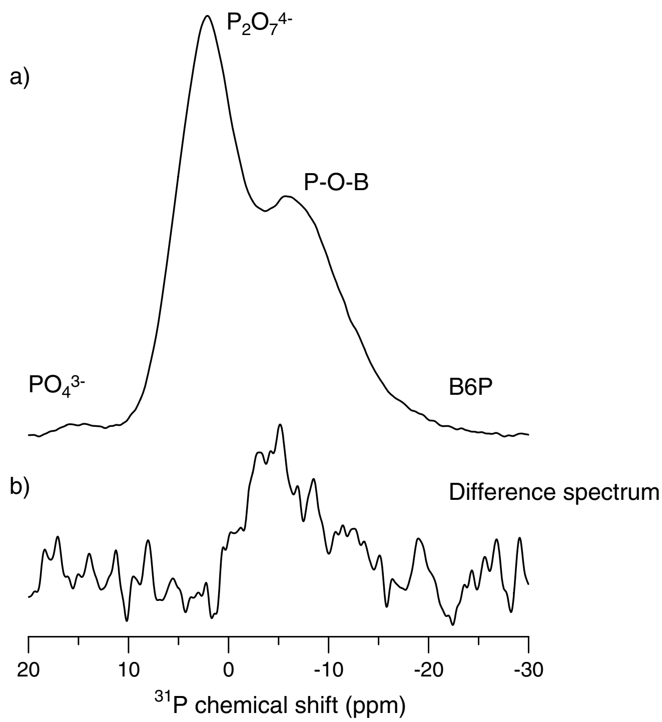

Fig. 5. ^{31}P–^{11}B TRAPDOR spectra of the B6P glass: ^{31}P MAS NMR spectrum (a) and the difference spectrum between the spectra with and without the ^{31}P–^{11}B transfer of population (b).

spectrum) with the difference between the MAS spectrum and the spectrum attenuated by the reintroduction of dipolar interaction. This difference spectrum shows which phosphate sites are connected to borate sites. Fig. 5 confirms that the broad resonance centered on −7 ppm stems from phosphates connected to borate groups, and that the diphosphate units at 3 ppm are not connected to borate groups. Concerning the monophosphates, at 15 ppm, their low intensity prevented a significant response about their possible connectivity with borates.

3.3. ^{31}P NMR chemical shielding calculations

Fig. 6 presents the molecular clusters used for the calculation of the ^{31}P NMR chemical shielding tensors. Cluster P1 ($H_4P_2O_7$) represents a diphosphate unit. Termination of the dandling bond with hydrogen allows keeping the charge neutrality during the calculation. Special care has been taken during the geometry optimization to impose symmetry constraints to ensure that both phosphorus atoms of the $P_2O_7^{4-}$ anion have the same shielding tensor. Cluster P1–B1 ($H_5P_2BO_9$) is used as a model of a diphosphate bonded to a single boron atom in trigonal coordination. Only the $OB(OH)_2$ fragment is allowed to relax during optimization of the structure in order to avoid too important relaxation effects that would not be realistic since the remaining glass network is not included in the calculation. With this procedure, only the electronic effect of the boron is taken into account in the calculation of the shielding tensor. The same partial optimization procedure has been used for all the clusters. Clusters P1–B2 and P1–B3 ($H_6P_2B_2O_{11}$) are used to model diphosphate groups bonded to two boron trigonal groups through the same phosphorus atom and through different phosphorus atoms, respectively. Cluster P1–B4 ($H_6P_2BO_{10}^-$) represents a diphosphate bonded to a tetrahedral boro-hydroxide group. In this case, the tetrahedral coordination of the boron atom imposes that the cluster carries a negative charge. In order to check the influence of the different cluster charge on the chemical shift calculations, we have built the cluster P1–B5, where a sodium atom has been added as a charge compensator of the tetragonal borate unit. Finally, clusters P1–S1 ($H_6P_2SiO_{10}$) and P1–S2 ($H_8P_2Si_2O_{13}$) were used to evaluate the chemical shift of phosphorus bonded to one or two silicon groups, respectively.

Isotropic chemical shieldings (σ) are reported in Table 2 along with chemical shifts computed using the standard formula ($\delta = \sigma_{ref} - \sigma$) for the two phosphorus atoms in each cluster model. σ_{ref} has been adjusted so that the chemical shift of the two phosphorus atoms of cluster P1 are equal to the experimental value (∼3 ppm). The chemical shift of phosphorus clearly decreases when trigonal boron is bonded to one or more oxygens of the PO_4 tetrahedra, but without affecting the chemical shift of the adjacent phosphorus atom in the diphosphate group (clusters P1–B1 and P1–B2). If tetrahedral boron is bonded to PO_4 tetrahedra, the phosphorus chemical shift also decreases, but that of the adjacent phosphorus in the diphosphate group is also affected. If silicon is bonded to phosphates, it also produces a decrease of the ^{31}P chemical shift, though less pronounced (clusters P1–Si1 and P1–Si2).

3.4. ^{29}Si, ^{11}B and ^{23}Na NMR

Fig. 7 presents the ^{29}Si MAS NMR spectra of the B, B3P and B6P glasses. The spectra of the phosphorus-containing glasses, B3P and B6P, show resonances centered on more negative chemical shifts. This lower ^{29}Si chemical shift indicates that the degree of polymerization of the silicate network increases, due to greater proportions of Q^3 and Q^4-type tetrahedra, or that the average quantity of BO_3 and BO_4 groups connected to silicates decreases.

Fig. 8 shows the ^{11}B MAS NMR spectra of two borosilicate glass compositions, B and D, and their corresponding phosphorus-containing glasses, B3P and D3P. When adding P_2O_5 in borosilicate glasses, no significant changes could be observed in the shape of the spectra. The ^{11}B MAS NMR spectra have been decomposed into four components [39], two of them corresponding to the ring and non-ring tri-coordinated boron atoms, and the two others attributed to the tetra-coordinated boron atoms, [BO_4], in (1B,3Si) and (0B,4Si) configurations [39]. It is worth to mention that in glasses with either high R or low K values there was no evidence of the peak at ∼−2 ppm assigned to [BO_4] in (0B,4Si) form.

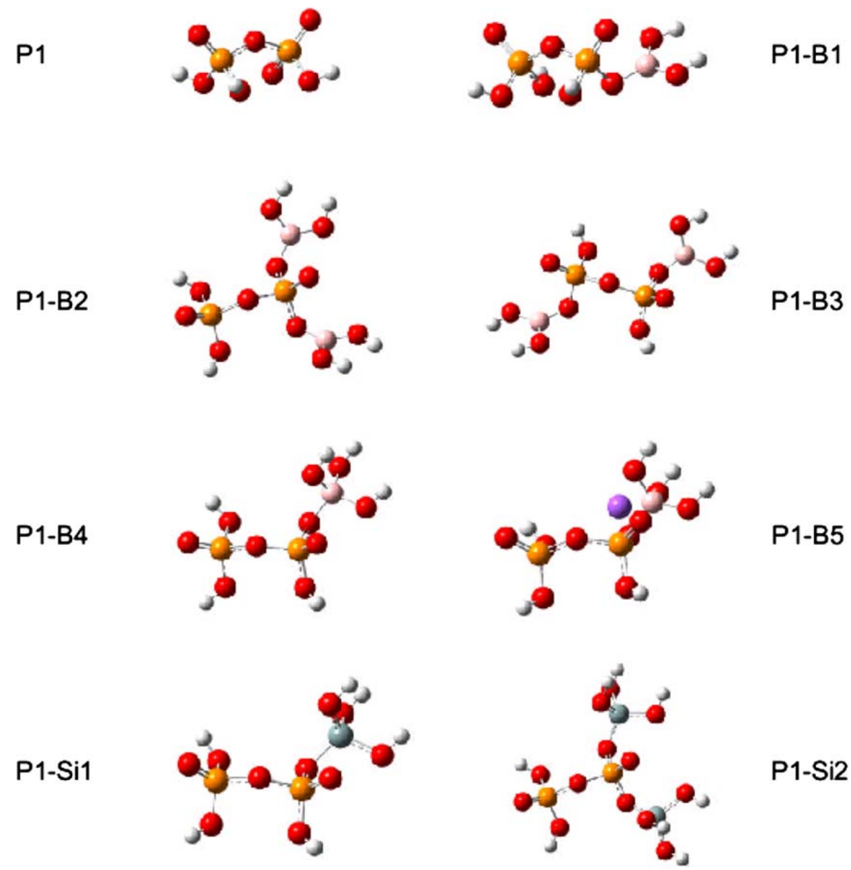

Fig. 6. Optimized configurations of the molecular clusters used for the calculation of the ^{31}P NMR chemical shielding tensors.

Table 2
Isotropic chemical shielding (σ) and chemical shifts ($\delta = \sigma_{ref} - \sigma$) of the two phosphorus sites in diphosphate groups

Model cluster	^{31}P isotropic chemical shielding (ppm) (P_{left}; P_{right})	δ (ppm)
P1	309.4; 309.4	3; 3
P1–B1	309.6; 315.6	2.8; −3.2
P1–B2	309.3; 321.4	3.1; −9
P1–B3	315.8; 315.7	−3.4; −3.3
P1–B4	308.9; 310.5	3.5; 1.9
P1–B5	312.7; 315.7	−0.3; −3.3
P1–Si1	309.7; 313.7	2.7; −1.3
P1–Si2	309.5; 317.8	2.9; −5.4

σ_{ref} has been adjusted so that the chemical shift of the two phosphorus atoms of cluster P1 are equal to the experimental value (3 ppm).

Fig. 9 shows the ^{23}Na MAS NMR spectra of glass E, and of the corresponding glass with 3 mol% P_2O_5-containing glass, E3P. The spectra significantly differ in chemical shift due to the different environment of Na^+ cations. The ^{23}Na chemical shift (δ) is correlated to the sodium–oxygen distance, as was established by Maekawa et al. [40]: $d_{Na-O} = -0.0119\delta + 2.5912$. A longer Na–O distance is expected when sodium acts as a charge compensating ion for BO_4 tetrahedra rather than when it is located on non-bridging oxygens (Si–O···Na) [41]. A decrease of the ^{23}Na chemical shift (i.e. a shift to more negative values)

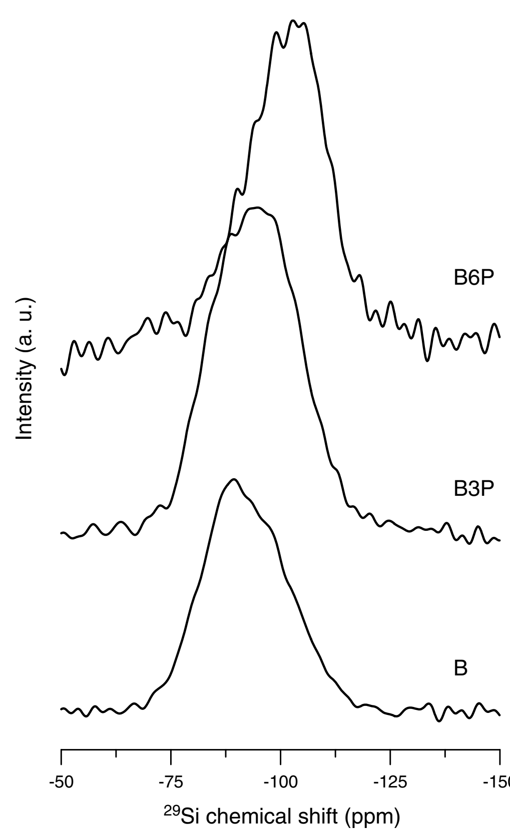

Fig. 7. ^{29}Si MAS NMR spectra of the B, B3P and B6P glasses.

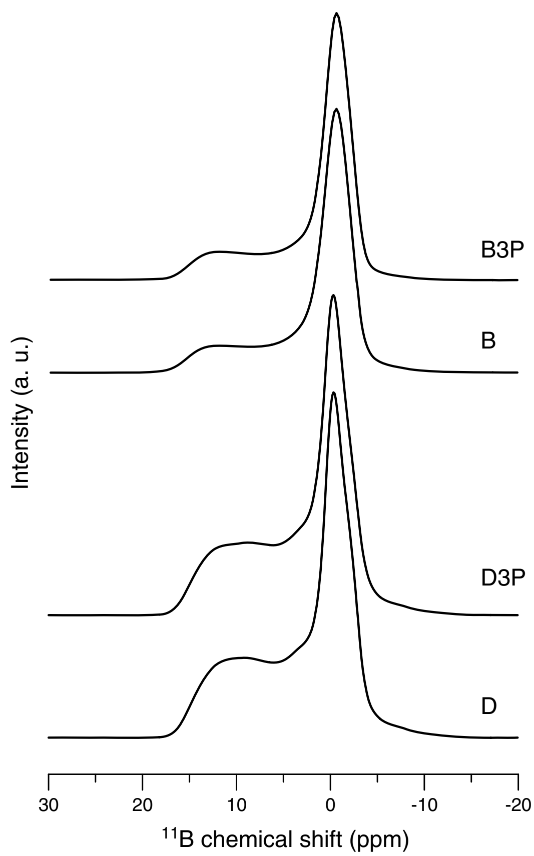

Fig. 8. ^{11}B MAS NMR spectra of, B, B3P, D and D3P glasses.

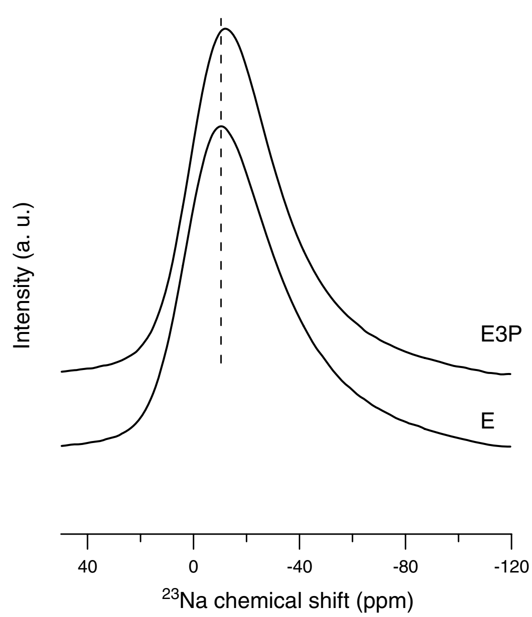

Fig. 9. ^{23}Na MAS NMR spectra of D, D3P, E and E3P glasses.

thus indicates that Na^+ ions move from $[BO_4]$ charge compensating position to non-bridging position [41]. By considering the composition of the glass E: $1.5Na_2O \cdot B_2O_3 \cdot 2SiO_2$, it can be deduced that both Na^+ cations coordinated with non-bridging oxygens of SiO_4 tetrahedra

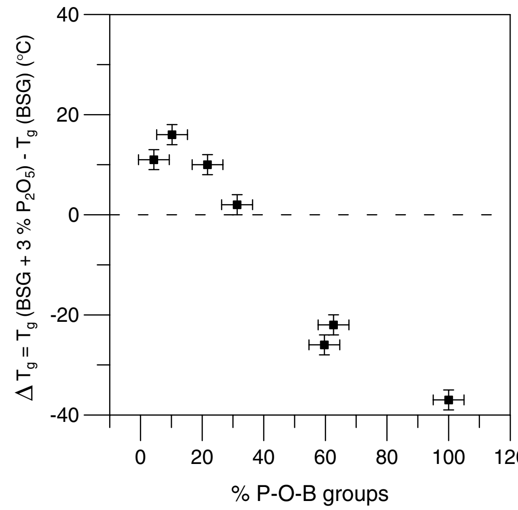

Fig. 10. Temperature difference between the T_g of the borosilicate glasses (BSG) with and without 3 mol% P_2O_5.

and charge compensating ones are expected in glass E. When introducing P_2O_5, the spectrum is slightly shifted to more negative values, meaning that part of the sodium ions turns to be coordinated with non-bridging oxygens of the phosphate tetrahedra.

3.5. Effect of the addition of P_2O_5 on the glass transition temperature

Fig. 10 reports the difference between the T_g of the 3 mol% P_2O_5-containing glasses and that of the borosilicate reference glass, plotted as a function of the P–O–B percentage measured by the de-convolution of ^{31}P NMR spectra. When the addition of P_2O_5 produces less than 38% of P–O–B groups, an increase in T_g is observed, compared to the glass without P_2O_5. On the contrary, a decrease in T_g is obtained when the addition of P_2O_5 results in more than 38% of P–O–B species.

4. Discussion

4.1. Assignment of resonances from the ab initio calculations

Table 2 shows that the ^{31}P chemical shifts of the phosphate groups bonded to borate groups in the cluster models have more negative values than those of phosphate groups non-bonded to borates. Nevertheless, boron does not affect the chemical shift of the second PO_4 group of the diphosphate cluster. Even in the case of a $[BO_4]$ group, either charge compensated or not with a sodium cation, a similar effect on the phosphorus chemical shift is observed, though in a lower extent. So it is confirmed from these first principle calculations that boron induces a decrease in the chemical shift of phosphates groups, which confirms the assignment to P–O–B groups of the resonance centered on −7 ppm.

The ^{31}P shielding tensors of diphosphates with P–O–Si bonds have also been calculated. Table 2 shows again a decrease in the chemical shift of the phosphorus bonded to silicon, but the difference with the value of the pure diphosphate cluster is lower than that with P–O–B bonds. Cody et al. [24] pointed out that in $Na_2O–Al_2O_3–SiO_2–P_2O_5$ glasses with a silica content of 88 mol%, phosphorus can form P–O–Si bonds for low Al_2O_3 contents. From *ab initio* calculations, they obtained a chemical shift value of 9 ppm for phosphate bonded to silicate, between the chemical shift values assigned to the mono- and diphosphate groups, i.e. 14 and 2 ppm, respectively. They indeed observed experimentally a small resonance around 10 ppm, which was attributed to phosphates connected to silicate through P–O–Si bonds. However, NMR investigations of $Na_2O–SiO_2–P_2O_5$ glasses showed a very different behavior [13]. The ^{31}P NMR spectra showed only two resonance bands, assigned to both the mono- and diphosphate species at 15 and 3 ppm, respectively, without any evidence for P–O–Si bonding or any other resonance than those of mono- and diphosphates. Since our borosilicate glasses contain a significant amount of Na_2O, we conclude, in accordance with reference [13], that the phosphates in our glasses are not connected to silicate groups.

The broad P–O–B resonance spread between −3 and −9 ppm suggests that there is more than one kind of phosphate involved. The $^{31}P–^{11}B$ TRAPDOR experiment shows that all resonances between 0 and −15 ppm are indeed associated with P–O–B bonds. However, the *ab initio* calculations reported in Table 2 indicate that diphosphate groups connected to both silicate and borate units may have a chemical shift in this region. Considering again the absence of evidence for P–O–Si connection in silicate glasses [13], we conclude from our *ab initio* calculations that the diphosphates can be connected to either one or two borates groups, which can be either tricoordinated or tetracoordinated borates.

4.2. Phosphate speciation as a function of the optical basicity of the borosilicate glasses, and its effect on T_g

The variation of the quantity of phosphate species with the glass composition is related to the optical basicity of the borosilicate glass, as shown in Fig. 11. As the glass basicity increases, a decrease of the P–O–B proportions and an increase of the proportions of the mono- and diphosphate units is observed. The speciation of P_2O_5 in borosilicate glasses may be explained in terms of the optical basicity Λ_g, with the following equilibriums:

$$P_2O_5 + 3O^{2-} \leftrightarrows P_2O_7^{4-} + O^{2-} \leftrightarrows 2PO_4^{3-} \quad (1)$$

Increasing the Na_2O content leads to a higher oxygen anion activity in the glass. According to Eq. (1), this favors the formation of less polymerized mono- and diphosphate species. The depolymerization of P_2O_5 into mono- and diphosphate units means that oxygens belonging to the network will have to be reorganized in order to enable the charge compensation of the phosphate units. As already demonstrated in $Na_2O–SiO_2–P_2O_5$ and $K_2O–B_2O_3–SiO_2–P_2O_5$ glasses [13,23,25], the incorporation of phosphorus into the glass induces a polymerization of the network, which can be written as:

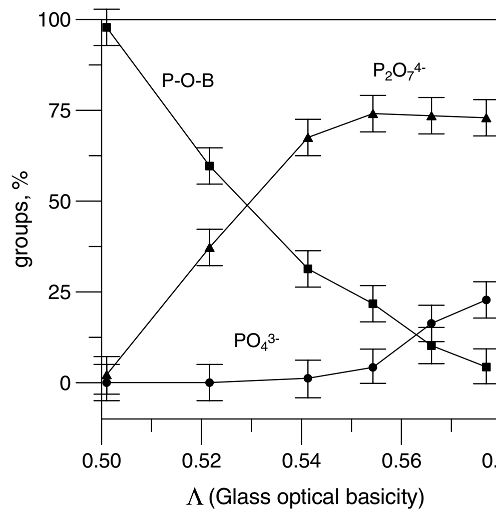

Fig. 11. Variation of the PO_4^{3-}, $P_2O_7^{4-}$ and P–O–B groups with the optical basicity of the borosilicate parent glasses, obtained from the ^{31}P NMR spectra fits. The lines are drawn as a guide to the eye.

$$2SiO^- \leftrightarrows Si–O–Si + O^{2-} \quad (2)$$

As shown in Fig. 7, the increase in the phosphorus content indeed produces an increase in the degree of polymerization of the glass network. The change produced in the T_g by the introduction of P_2O_5 in the borosilicate glasses (Fig. 10) can be explained in terms of polymerization of the borosilicate matrix. In glasses with a large basicity index Λ_g, i.e. a high Na_2O content, the formation of mono- and diphosphate species takes place preferentially with the Na^+ ions belonging to the silicate network and giving rise to the subsequent polymerization through Eq. (2). If not enough Na^+ ions are available, phosphate groups will tend to form P–O–B bonds with the borate groups. If many P–O–B are formed, less sodium ions will leave the silicate network, meaning that the polymerization effect will be limited. So, the increase in the glass network polymerization will be determined by the quantity of Na_2O that is involved in the formation of sodium phosphate groups. Thus, the higher the increase in the degree of polymerization of the silicate network, the higher the increase in the glass transition temperature (Fig. 10). As more P–O–B bonds are formed, the increase in the degree of polymerization of the silicate network turns to be moderated since less Na^+ cations become involved in the formation of phosphate groups, then giving rise to a smaller increase, or even a decrease, in the T_g of the P_2O_5-containing glass compared to that of the borosilicate one (above 38% P–O–B). On the contrary, if ortho- and pyrophosphates appear as predominant species over the P–O–B bonds (below 38% P–O–B), then the silicate re-polymerization induces an increase in T_g. However, it is worth mentioning that results for compositions with a higher tendency to present phase separation, i.e. very low Na_2O containing glasses, or those which easily form phosphate compounds by crystallization, i.e. high Na_2O content, must be carefully taken into account while interpreting properties dependency with composition.

Figs. 12 and 13 present the variation of the three phosphate species in borosilicate glasses with 3 mol% of P_2O_5 as a function of K ($R = 1$) and R ($K = 2$), respectively. When varying K from 0.86 to 2, an increase in the P–O–B proportion is observed together with a decrease in the PO_4^{3-} and $P_2O_7^{4-}$ ones. However, when R varies from 0.5 to 1.5, the P–O–B content progressively decreases while those of PO_4^{3-} and $P_2O_7^{4-}$ units increase. Therefore, P–O–B bonds are formed when either SiO_2 content increases for constant alkali content, or when Na_2O decreases for a constant SiO_2/B_2O_3 ratio. This evolution can be explained by considering that when P_2O_5 is added in a glass with a large SiO_2-content, it will compete with borate for the Na^+ charge compensator ions. If there is not enough Na^+ available, phosphates will tend to form P–O–B groups in order to saturate their non-bridging oxygens. Fig. 14 depicts the percentages of the three kinds of phosphate species in the $Na_2O \cdot B_2O_3 \cdot 1.3SiO_2$ (A) and $Na_2O \cdot B_2O_3 \cdot 2SiO_2$ (B) glasses as a function of the P_2O_5 content. The proportion of the P–O–B bonds increases while the mono- and diphosphate ones decrease as phosphorus increases in the glasses. So again, there is a tendency to form a glass network in which phosphorus will preferentially form P–O–B species.

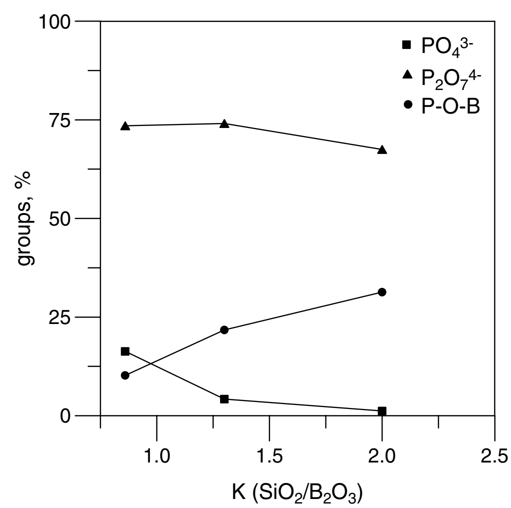

Fig. 12. Variation of the phosphate groups in borosilicate glasses for a constant 3 mol% of P_2O_5 as a function of K ($R = 1$). The lines are drawn as a guide to the eye.

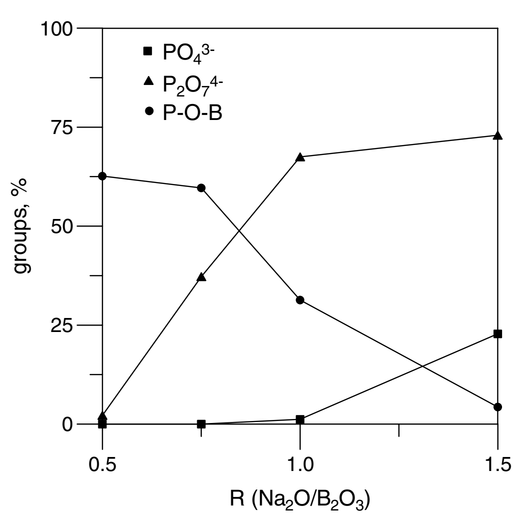

Fig. 13. Variation of the phosphate species in borosilicate glasses for a constant 3 mol% of P_2O_5 as a function of R ($K = 2$). The lines are drawn as a guide to the eye.

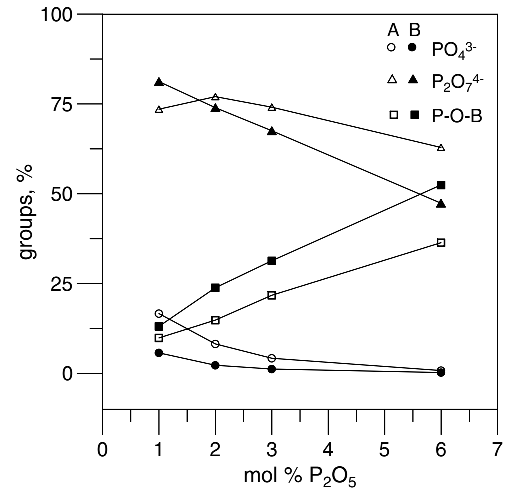

Fig. 14. Variation of the phosphate species in A and B glasses, as a function of the P_2O_5 content. The lines are drawn as a guide to the eye.

5. Conclusions

The introduction of P_2O_5 in borosilicate glasses leads to the formation of isolated PO_4^{3-}, $P_2O_7^{4-}$ anions, and to $P_2O_7^{4-}$ anions connected to tricoordinated and tetracoordinated borate groups. ^{31}P MAS NMR spectra showed three distinct resonances for these phosphate species. ^{31}P–^{11}B TRAPDOR NMR confirmed that phosphates are connected to borates, and ^{31}P DQ NMR showed that only $P_2O_7^{4-}$ groups are connected to borates. These experimental features were confirmed by *ab initio* chemical shift calculations, which also suggested that some bonding of phosphate groups to silicate tetrahedra may occur, but it was excluded considering that they do not appear in silicate glasses.

The relative quantity of PO_4^{3-}, $P_2O_7^{4-}$ and to $P_2O_7^{4-}$ anions connected to borate groups is related to the optical basicity of the borosilicate glasses, and hence to the quantity of sodium ions that can be extracted from the borosilicate network for the charge compensation of the phosphate anions. The evolution of the glass transition temperature with the glass compositions can be explained according to this structural model.

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References

[1] M.J. Pascual, A. Durán, L. Pascual, Phys. Chem. Glasses 43 (2002) 25.
[2] M.J. Pascual, A. Durán, L. Pascual, J. Non-Cryst. Solids 306 (2002) 58.
[3] K. Tajima, M. Ohashi, K. Shiraki, M. Tateda, S. Shibata, J. Lightwave Technol. 10 (1992) 1532.
[4] E.M. Dianov, M.V. Grekov, I.A. Bufetov, S.A. Vasiliev, O.I. Medvedkov, V.G. Plotnichenko, V.V. Koltashev, A.V. Belov, M.M. Bubnov, S.L. Semjonov, A.M. Prokhorov, Electron. Lett. 33 (1997) 1542.
[5] E.M. Dianov, I.A. Bufetov, M.M. Bubnov, M.V. Grekov, S.A. Vasiliev, O.I. Medvedkov, Opt. Lett. 25 (2000) 402.
[6] C. Canevali, R. Scotti, A. Vedda, M. Mattoni, F. Morazzoni L. Armelao, D. Barreca, G. Bottaro, Chem. Mater. 16 (2004) 315.
[7] M. Fanciulli, E. Bonera, E. Carollo, L. Zanotti, Microelectron. Eng. 55 (2001) 65.
[8] M. Nogami, Y. Daiko, T. Akai, T. Kasuga, J. Phys. Chem. B 105 (2001) 4653.
[9] Y. Daiko, T. Akai, T. Kasuga, M. Nogami, Chem. Mater. 14 (2004) 4624.
[10] F.J. Ryerson, P.C. Hess, Geochem. Cosmochem. Acta 42 (1978) 921.
[11] M.J. Toplis, D.B. Dingwell, G. Libourel, Contrib. Mineral. Petrol. 117 (1994) 293.
[12] H. Grussaute, L. Montagne, G. Palavit, J.L. Bernard, Glastech. Ber. Glass Sci. Technol. 73 (2000) 380.
[13] H. Grussaute, PhD thesis, Université des Sciences et Technologies de Lille, 1998.
[14] H. Grussaute, L. Montagne, G. Palavit, J.L. Bernard, J. Non-Cryst. Solids 263&264 (2000) 312.
[15] W. Vogel, Glass Chemistry, Springer, Berlin, 1985.
[16] W.L. Gong, L.M. Wang, R.C. Ewing, E. Vernaz, J.K. Bates, W.L. Ebert, J. Nucl. Mater. 254 (1998) 249.
[17] S.Z. Xiao, J. Non-Cryst. Solids 45 (1981) 29.
[18] J. Zhong, P.J. Bray, J. Non-Cryst. Solids 111 (1989) 67.
[19] B.C. Bunker, D.R. Tallant, R.J. Kirkpatrick, G.L. Turner, Phys. Chem. Glasses 31 (1990) 30.
[20] W.J. Dell, P.J. Bray, S.Z. Xiao, J. Non-Cryst. Solids 58 (1983) 1.
[21] R. Martens, W. Müller-Warmuth, J. Non-Cryst. Solids 265 (2000) 167.
[22] C. Rong, K.C. Wong-Moon, H. Li, P. Hrma, H. Cho, J. Non-Cryst. Solids 223 (1998) 32.
[23] H. Gan, P.C. Hess, R.J. Kirkpatrick, Geochim. Cosmochim. Acta 58 (1994) 4633.
[24] G.D. Cody, B. Mysen, G. Sághi-Szabó, A. Tossell, Geochim. Cosmochim. Acta 65 (2001) 2395.
[25] R. Dupree, D. Holland, M.G. Mortuzza, J.A. Collins, M.G.W. Lockyer, J. Non-Cryst. Solids 112 (1989) 111.
[26] R. Dupree, D. Holland, M.G. Mortuzza, Nature 328 (1987) 416.
[27] M. Feike, R. Graf, I. Schnell, C. Jäger, H.W. Speiss, J. Am. Chem. Soc. 118 (1996) 9631.
[28] K.K. Olsen, J.W. Zwanziger, P. Hartmann, C. Jäger, J. Non-Cryst. Solids 222 (1997) 199.
[29] R. Ernst, G. Bodenhausen, A. Wokaun, Principles of Nuclear Magnetic Resonance in One and Two Dimensions, Clarendon, Oxford, 1987.
[30] D. Massiot, F. Fayon, M. Capron, I. King, S. Le Calvé, B. Alonso, J.O. Durand, B. Bujoli, Z. Gan, G. Hoatson, Magn. Reson. Chem. 40 (2002) 70.
[31] M.J. Frisch, G.W. Trucks, H.B. Schlegel, G.E. Scuseria, M.A. Robb, J.R. Cheeseman, J.A. Montgomery Jr., T. Vreven, K.N. Kudin, J.C. Burant, J.M. Millam, S.S. Iyengar, J. Tomasi, V. Barone, B. Mennucci, M. Cossi, G. Scalmani, N. Rega, G.A. Petersson, H. Nakatsuji, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, M. Klene, X. Li, J.E. Knox, H.P. Hratchian, J.B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R. Gomperts, R.E. Stratmann, O. Yazyev, A.J. Austin, R. Cammi, C. Pomelli, J.W. Ochterski, P.Y. Ayala, K. Morokuma, G.A. Voth, P. Salvador, J.J. Dannenberg, V.G. Zakrzewski, S. Dapprich, A.D. Daniels, M.C. Strain, O. Farkas, D.K. Malick, A.D. Rabuck, K. Raghavachari, J.B. Foresman, J.V. Ortiz, Q. Cui, A.G. Baboul, S. Clifford, J. Cioslowski, B.B. Stefanov, G. Liu, A. Liashenko, P. Piskorz, I. Komaromi, R.L. Martin, D.J. Fox, T. Keith, M.A. Al-Laham, C.Y. Peng, A. Nanayakkara, M. Challacombe, P.M.W. Gill, B. Johnson, W. Chen, M.W. Wong, C. Gonzalez, J. Pople, Gaussian 03, Revision B.05, Gaussian, Inc., Wallingford, CT, 2004.
[32] A.D. Becke, J. Chem. Phys. 98 (1993) 5648.
[33] K. Wolinski, J.F. Hilton, P. Pulay, J. Am. Chem. Soc. 112 (1990) 8251.
[34] W. Kutzelnigg, U. Fleischer, M. Schindler, NMR, Basic Principles and Progress, vol. 23, Springer, Berlin, 1990, p. 165.
[35] J.A. Duffy, M.D. Ingram, J. Non-Cryst. Solids 21 (1976) 373.
[36] R.J. Kirkpatrick, R.K. Brow, Solid State Nucl. Magn. Reson. 5 (1995) 9.
[37] R. Witter, P. Hartmann, J. Vogel, C. Jäger, Solid State Nucl. Magn. Res. 13 (1998) 189.
[38] S. Donze, L. Montagne, G. Palavit, C. Jaeger, M. Zeyer, J. Non-Cryst. Solids 263&264 (2000) 132.
[39] L.-S. Du, J.F. Stebbins, J. Non-Cryst. Solids 315 (2003) 239.
[40] H. Maekawa, T. Nakao, S. Shimokawa, T. Yookawa, Phys. Chem. Miner. 24 (1997) 53.
[41] F. Angeli, J.M. Delaye, T. Charpentier, J. Non-Cryst. Solids 276 (2000) 132.